

The potential role of curcumin in patients with monoclonal gammopathy of undefined significance--its effect on paraproteinemia and the urinary N-te...



PURPOSE: To determine the effect of curcumin on plasma cells and osteoclasts in patients with MGUS. **EXPERIMENTAL DESIGN:** Twenty-six patients with MGUS were recruited into the study and administered 4 grams/day oral curcumin. Blood and urine samples were collected at specified visits after initiating therapy. Full blood count, B2 microglobulin, serum paraprotein, and immunoglobulin electrophoresis (IEPG and EPG) were determined for all patients at each visit. Serum calcium, 25 hydroxyvitamin D3, and bone-specific alkaline phosphatase were determined at baseline only. Urine, as a morning second-void sample, was collected at each visit for urinary N-telopeptide of type I collagen. **RESULTS:** Our results show that oral curcumin is able to decrease paraprotein load in a select group (i.e., those having a paraprotein level of >20 g/L) of patients with MGUS. Fifty percent (5 of 10) of these patients had a 12% to 30% reduction in their paraprotein levels, while on curcumin therapy. In addition, 27% of patients on curcumin had a >25% decrease in urinary N-telopeptide of type I collagen. **CONCLUSION:** Due to the possible progression of MGUS to multiple myeloma, the potential role of curcumin as a therapeutic intervention for MGUS patients warrants further investigation.